

connectivity to the equipment in the network. The major advantage of LAN Ethernet port is the complete support for remote control of the instrument including viewing of waveform data and transfer of the data and set-up files from any location. LAN also leverages the Internet, allowing you to access the equipment in India from anywhere in the world."

Explaining the way it works, Bhatia shares, "LXI instruments have a Web server on the equipment itself. Once you connect it in the network, an IP is assigned to it. You can open any Web browser like Internet Explorer to control the oscilloscope in your network." The benefit is that all the equipment are interconnected and you can plan an event based on triggers from other equipment and monitor them remotely.

You can also file transfer protocol (FTP) into a digital storage oscilloscope (DSO) in the network to grab the data from its memory.

Virtual instrumentation for customised scopes

As devices become more complex and test times costlier, software-defined instruments offer significant flexibility, integration and cost benefits over traditional architectures based on standalone box instruments. "With digitisers driven by flexible software, test engineers can perform standard oscilloscope measurements and easily build other instruments such as spectrum analysers, transient recorders and ultrasonic receivers," says Subramanya. Sometimes there is a need of tight synchronisation within the instruments at picoseconds-level accuracy for high-channel-count and mixed-signal applications.

"Replacement of oscilloscopes is not inevitable as these instruments have become reputable solutions for testing, debugging and troubleshooting electrical signals. But with the advent of virtual instrumentation, we may come across new instruments that combine the capabilities of oscil-

loscope with other test equipment as an all-in-one solution," shares Narasimhan.

Due to increasing product complexity, shorter times to market, increasing hardware and software abstraction, and reduced budgets, monolithic instrumentation architectures with predefined functionality are unable to meet the current demands.

Subramanya says, "Virtual instrumentation technology has pioneered the concept of user-defined systems to design, prototype and deploy test, control and embedded applications. This strong framework which leverages graphical programming software and modular hardware has helped customers to simplify development, increase productivity and dramatically reduce time to market."

Talking about modularity, Subramanya shares, "PXI provides a rugged PC-based platform that continues to offer the highest bandwidth and lowest latency. PXI digitisers are optimised for automated test because of their high data throughput, tight synchronisation between channels and ease of integration with other types of instrumentation."

Oscilloscopes are available in the form of high-speed digitisers. "Digitiser cards have their own attractiveness if users need to combine a few instruments for repeated set of customised testing or for higher vertical bit resolution. Customisation also means the need to invest on software programming efforts. Digitiser cards can replace an oscilloscope in system level to a certain extent if the user uses it just as an ADC to fetch data," says Appala Srinivasa Rao, area manager-product support and applications, Rohde & Schwarz.

Application software: Key to better user experience

The whole idea of PC connectivity will be bullied if you do not have a good application software. Narasimhan shares, "Many oscilloscope applications need special software applica-

tions that make it easier for the user to analyse and evaluate the measurements in an automated way."

Vendors see a growing need in this area, particularly for mid-range products that can provide performance measurements at an affordable cost. Narasimhan adds, "Newer applications such as automated power measurements, jitter analysis, and low- and medium-speed serial bus decoding are being integrated in the oscilloscope software." To give a near-PC experience, oscilloscopes have Microsoft Windows—the most common operating system—running beneath.

"With specialised application software functions, users can download the image of a waveform with measurable parameters. It provides the comma-separated variables (CSV) file format so that data points of the sampled signal can be collected to make detailed analysis of the signal like Fourier's Analysis," shares Kumar.

Challenges in PC-based scopes

As we talk about the power of PC, it is important to understand the disadvantages as well:

1. Unlike standalone box equipment, you need to install oscilloscope software on your computer to operate the equipment from the computer.
2. The PC will take its own defined time to boot, compared to the instant start-up of a self-contained oscilloscope. However, this distinction is narrowing as the latest scopes too come with a full-fledged operating system like Windows, which may take some time to boot.
3. Multi-threading and multi-tasking on the PC can be a distraction in oscilloscope operation.
4. Oscilloscopes last longer than PCs and hence it may happen that the scope is running an older version of the operating system with weaker protection against virus. If the scope is affected by the virus attack, it may spread the virus throughout the network.